

Introduction to Machine Learning

Learning Outcomes

After completing this lecture, students will be able to:

- Define Machine Learning (ML)
- Understand real-world applications of ML
- Differentiate Classification and Regression
- Explain Learning Associations
- Understand Unsupervised Learning
- Explain Reinforcement Learning

What is Machine Learning?

Definition

Machine Learning (ML) is a branch of Artificial Intelligence (AI) that enables computers to learn from data and improve their performance without being explicitly programmed.

Key Idea

Machine Learning uses data, algorithms, and experience to make predictions or decisions.

Goal:

- Extract useful information from data
- Build predictive models
- Automate decision-making processes

Traditional Programming vs Machine Learning

In traditional programming, a programmer explicitly writes rules to solve a problem.

$$\text{Input} + \text{Program} \rightarrow \text{Output}$$

Example:

Suppose we want to determine whether a student has passed an examination or not.

- Input: Marks = 95
- Program Rule: If marks > 50 , then Pass
- Output: Pass

The computer follows the predefined rules written by the programmer.

Traditional Programming vs Machine Learning

Machine Learning

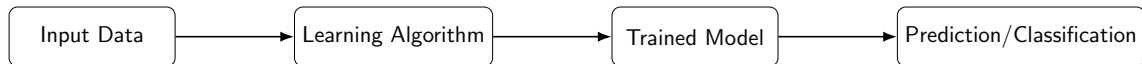
In Machine Learning, instead of writing explicit rules, we provide examples (data) from which the machine learns patterns automatically.

Model Training

Input Data + Output Data $\rightarrow$ Learning Algorithm $\rightarrow$ Model

Model Testing

New Input $\rightarrow$ Model $\rightarrow$ Prediction/Classification

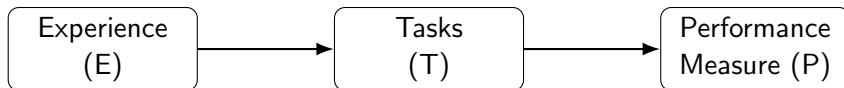


Data + Learning Algorithm = Intelligent Model

Formal Definition of Machine Learning

Mitchell's Definition of Machine Learning

A computer program is said to learn from **Experience (E)** with respect to some class of **Tasks (T)** and **Performance Measure (P)**, if its performance at tasks in T , as measured by P , improves with experience E .



Examples

- Experience (E): Training Data
- Tasks (T): Classification, Prediction, Control
- Performance (P): Accuracy, Error Rate, Reward

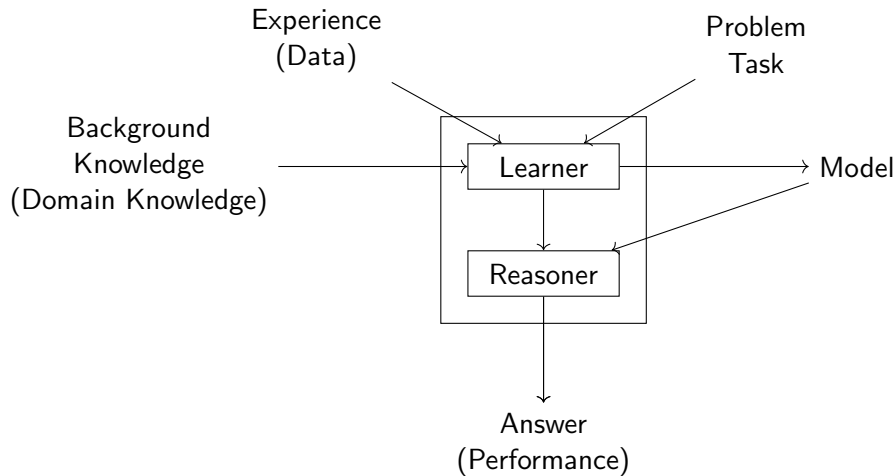
Example: Email Spam Detection

Term	Meaning
Task (T)	Classify emails as Spam or Not Spam
Experience (E)	Previously labeled emails
Performance (P)	Classification Accuracy

Training Examples

Email	Label
Win \$1000 Now!	Spam
Meeting at 2 PM	Not Spam
Claim Your Prize	Spam
Project Report Attached	Not Spam

Schematic Diagram of the ML Mode



For Creating a Learner

- 1 Choose the training experience (data/features).
- 2 Choose the target function
(that is to be learned).
- 3 Choose how to represent the target function
(Linear or Non-linear representation).

Examples of Machine Learning Applications

- Email Spam Detection
- Face Recognition
- Medical Diagnosis
- Credit Card Fraud Detection
- Speech Recognition
- Autonomous Vehicles
- Product Recommendation
- Weather Forecasting
- Stock Prediction
- Chatbots
- Image Classification
- Machine Translation

ML is used whenever systems learn patterns from data.

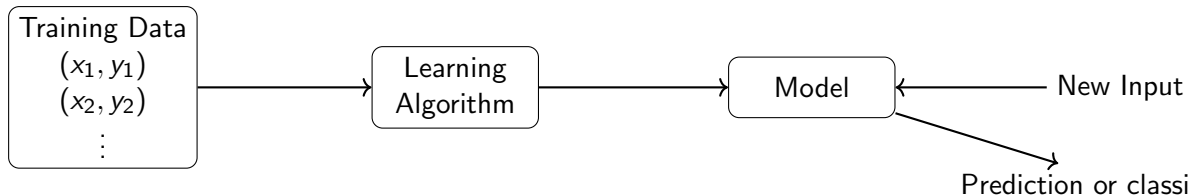
Types of Machine Learning

Supervised Learning

- Learning using labeled data
- Training data consists of input-output pairs (X, Y) .
- The learning algorithm learns a mapping:

$$f : X \rightarrow Y$$

- For a new input, the model predicts the corresponding output.



Goal: Learn from labeled examples and predict outputs for unseen data.

Training Examples in Supervised Learning

Training Dataset

	x_1	x_2	x_3	$\dots$	Y
Input ₁	a_1	a_2	a_3	$\dots$	y_1
Input ₂	b_1	b_2	b_3	$\dots$	y_2
Input ₃	c_1	c_2	c_3	$\dots$	y_3

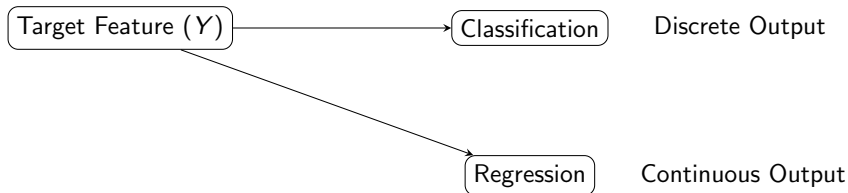
- $x_1, x_2, \dots, x_n$ are called **features** (input attributes).
- Y is the **target attribute** to be predicted.
- Each row represents one training example.

For a new instance

$$(z_1, z_2, \dots, z_n)$$

the model predicts the corresponding target value.

Target Feature in Supervised Learning



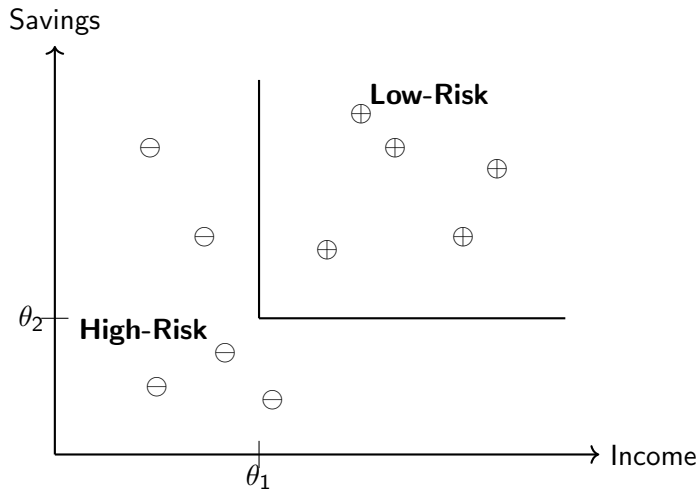
Classification

- Output belongs to a finite set of classes.
- Examples:
 - Rain / No Rain
 - Low-Risk / High-Risk
 - Spam / Not Spam

Regression

- Output is a continuous numerical value.
- Example: $\text{Price} = f(\text{Location}, \text{Floor Area}, \text{No. of BHKs})$

Credit Scoring as a Classification Problem



A discriminant separates customers into Low-Risk and High-Risk groups.

Credit Scoring as a Classification Problem

Learned Classification Rule

IF $\text{Income} > \theta_1$ AND $\text{Savings} > \theta_2$

THEN Low-Risk ELSE High-Risk

This rule acts as a **discriminant function** that separates customers into *Low-Risk* and *High-Risk* classes.

Regression in Supervised Learning

Regression Problem

- The output variable is a **continuous numerical value**.
- Example: Predicting the price of a used car.
- Input attributes may include:
 - Mileage
 - Price

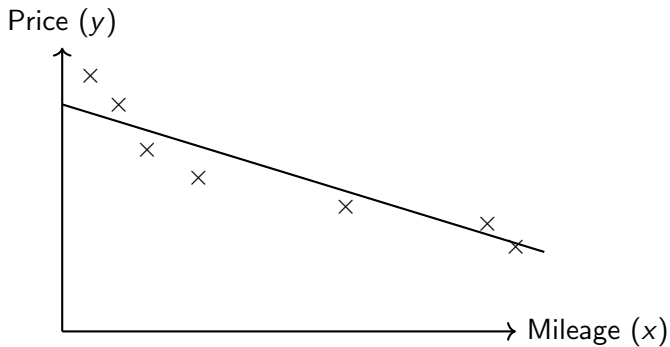
- **Regression Function**

The goal is to learn a function that predicts the car price from its mileage:

$$y = wx + w_0$$

where

- w = slope (weight)
- w_0 = intercept (bias)



A fitted line is the line that best matches the training data and is used to make predictions for new inputs.

Regression Model Learning

General Machine Learning Model

$$y = g(x \mid \theta)$$

- $g(\cdot)$: model/function
- θ : model parameters
- Goal: Find parameters that minimize prediction error and start with random initial values.

Model	Equ	Parameters (θ)
Linear Regression	$y = wx + w_0$	$\{w, w_0\}$
Polynomial Regression	$y = w_2x^2 + w_1x + w_0$	$\{w_2, w_1, w_0\}$
Neural Network	$y = f(w_1x_1 + w_2x_2 + b)$	All weights and biases

The learning algorithm adjusts the parameters to obtain the best fit to the training data and accurately predict the price of a new car.

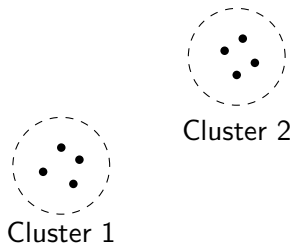
Unsupervised Learning

Definition

Unsupervised learning works with unlabeled data and discovers hidden patterns automatically.

Tasks:

- Clustering
- Dimensionality Reduction
- Association Discovery



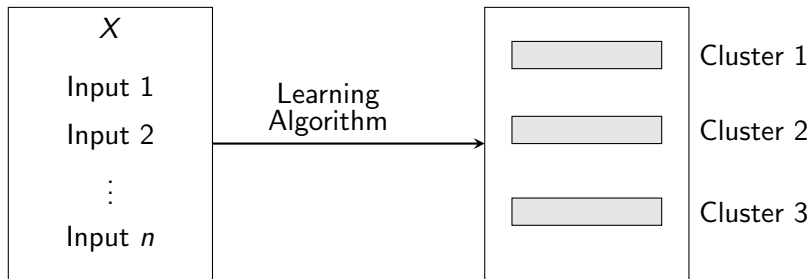
Definition

- Only input features (X) are available.
- No target/output variable (Y) is provided.
- The objective is to discover hidden patterns in data.
- The algorithm groups similar instances into clusters.

$$X = \{x_1, x_2, \dots, x_n\}$$

Given a set of features, cluster them or summarize them.

Unsupervised Learning: Clustering



How are clusters formed?

Based on similarity among data instances

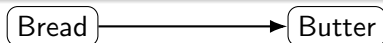
Learning Associations

Association Learning

Association learning discovers relationships among items that frequently occur together.

Example

Customers who buy Bread often buy Butter.



Association Rule

Applications:

- Market Basket Analysis
- Recommendation Systems
- Customer Behavior Analysis

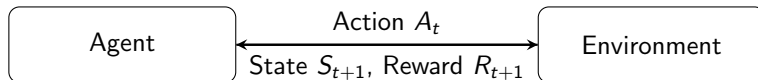
Definition

In Reinforcement Learning, an agent learns by interacting with the environment and receiving rewards or penalties.

Applications:

- Robotics
- Self-driving Cars
- Chess and Game Playing
- Resource Optimization

Reinforcement Learning



Learns a Policy π

Provides Feedback

- Agent interacts with the environment.
- Agent chooses an action based on the current state.
- Environment returns a new state and reward.
- Goal: Maximize cumulative reward over time.

Components of Reinforcement Learning

- **Agent:** Learner / decision maker (self-driving car)
- **Environment:** Road, traffic signals, other vehicles
- **States (S):** Current Situation of the environment
 - S_1 : Red Signal
 - S_2 : Green Signal
- **Actions (A):** Set of possible moves taken by the agent.
 - A_1 : Stop
 - A_2 : Go
- **Reward (R):** Feedback signal from the environment.
 - $R(S_1, A_1) = +10$ (Safe behaviour)
 - $R(S_1, A_2) = -50$ (Traffic Violation)
 - $R(S_2, A_2) = +50$ (Correct Action)
 - $R(S_2, A_1) = -5$ (Unnecessary Stop)
- **Policy (π):** Strategy used by the agent to select action overtime, the car learns the rule (Policy)

Comparison of Learning Types

Learning Type	Data	Output
Classification	Labeled	Category
Regression	Labeled	Numeric Value
Unsupervised	Unlabeled	Hidden Patterns
Reinforcement	Rewards	Optimal Action

Summary

- Machine Learning enables systems to learn from data.
- ML is widely used in healthcare, finance, and automation.
- Classification predicts categories.
- Regression predicts numerical values.
- Unsupervised Learning finds hidden structures.
- Reinforcement Learning learns through rewards.